

Diagnosing Late-Time Accuracy Limits of Single-Shot Acoustic Neural Operators in Heterogeneous Media

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Abstract

Fast acoustic wavefield prediction requires accuracy in the scattered coda as well as the first arrival. We study this requirement in a 39.1-million-parameter single-shot neural operator for heterogeneous two-dimensional media. Our contribution is an auditable diagnostic protocol combining time- and frequency-resolved error, correction-direction probes, and controlled intervention tests. On frozen training panels, full-future relative L^2 is 0.0609, 0.2303, and 0.3965 for uniform, layered, and Marmousi media, respectively. For Marmousi, the final time third contains 64.1% of the mean per-record error share but only 30.6% of the corresponding signal-energy share. None of its 96 records reaches 0.1. A preregistered 18,700-step intervention increases late-window error by 2.83%, despite reducing its training objective. Separate-frame adaptation probes indicate limited temporal transfer, while fixed-feature correction probes motivate direct residual supervision. At a separate validation endpoint, equalizing frequency-band errors to the lowest-band value would reduce pooled error by 23.5%, leaving 0.3641; this is a conditional counterfactual. A synchronized GPU benchmark measures 3.97s per query, 2.5–5.1 $\times$ faster than two classical configurations producing the same stored field, with higher prediction error. On train-development records, a coarse 51×51 classical solve shows the classical numerical-dispersion signature (phase lag, spectral downshift) where the operator’s late error is an in-phase amplitude deficit; that coarse solve is nevertheless more accurate than the operator on the layered record. A preregistered equal-budget DeepONet baseline collapsed during training, so no same-caliber baseline number exists. Translated Marmousi twins limit generalization claims, and the test split remains sealed. These results localize the measured bottleneck to late-time prediction and define targeted experiments in temporal coverage, residual learning, and solver-consistent correction.

1 Introduction

Acoustic simulation in heterogeneous media must recover both direct arrivals and the scattered wavefield. Neural operators amortise this mapping across repeated queries by learning from numerical solutions [1]. A direct, time-conditioned operator can query the entire future without feeding predictions back as inputs. This avoids autoregressive feedback, but it does not prevent approximation errors in amplitude or phase from increasing with the prediction horizon. Recovering the late coda therefore remains a separate accuracy requirement.

Our study asks what limits that accuracy in a hierarchical operator trained on uniform, layered, and Marmousi velocity models. The model combines medium and source encodings with a learned low-rank field and a spectral correction. The initial arrivals are reproduced more accurately than the late wavefield. Even on the training panel, Marmousi relative L^2 rises from 0.1753 in the first time third to 0.5728 in the last. An aggregate score alone does not reveal whether this deterioration reflects weak signal, missing frequency content, insufficient supervision, or an ineffective correction.

We address this ambiguity with a diagnostic protocol that connects each proposed intervention to a measurable error component. Exact energy decompositions separate signal energy

Table 1: Dataset. Attributes are read directly from the stored file.

Property	Value
Solver grid	401×401 at $\Delta x = \Delta z = 5.0$ m
Stored grid	201×201 at $\Delta x = \Delta z = 10.0$ m
Restriction	separable binomial-5 lowpass, then nodal decimation by 2
Frames per record	401, $\Delta t = 2.5$ ms, record length 1.0 s
Solver	LWC-84: 4th order in time, 8th order centred in space
Boundaries	unsplit three-sided CFS-CPML; node-centred free surface
Source	Ricker, $f_0 \in [8.0036, 29.9954]$ Hz over 4003 records
Splits	train 2800, validation 600, test_id 600 (sealed)
Families reported	uniform / layered / marmousi

from error mass. Conditional oracles quantify what an intervention could change while holding other components fixed. Controlled experiments then test specific optimizer, kernel, and correction settings. The identities are standard; the contribution is their combined use to prioritize experiments and document failed routes in this full-wavefield acoustic setting.

Three findings organize the paper. First, late Marmousi coda contains disproportionate error without a vanishing signal denominator. Second, the tested interventions and temporal-transfer probes narrow the remaining explanations without establishing a universal architectural floor. Third, the measured speed advantage has an explicit accuracy cost. Standard decoding takes 3.97 s per query against 10.00 and 20.07 s for two classical configurations on the same GPU and output grid. These findings provide an experimental basis for improving complex-media prediction; they do not establish that the accuracy target has been reached.

2 Setting

2.1 Problem and data

We solve the 2-D acoustic wave equation with constant density and variable velocity on a $2000 \text{ m} \times 2000 \text{ m}$ domain, with a free surface on top and absorbing conditions on the other three sides. The learning task is the single-shot operator

$$(8\text{-frame IC}, c(x, z), \text{source}) \mapsto p(x, z, t), \quad t \leq 1.0 \text{ s},$$

on the full stored grid. Table 1 gives the dataset, read from the HDF5 attributes.

The **anomaly** medium is held out of training and is not used anywhere here, so the validation evaluation covers 480 records (90 uniform, 240 layered, 150 marmousi).

2.2 Model

The anchor is checkpoint step 29359, a grouped UFNO with attention, MIONet fusion and a pyramid coarse field, feeding a dense wavefield decoder whose spectral kernels dominate the parameter count. A checkpoint-level audit gives a total of 39,103,589 parameters: velocity encoder 22,757,312, dense decoder 15,622,562 (spectral kernels alone 15,411,200), MIONet fusion 230,787. A configuration-level count in a separate artifact gives 39,104,869; the sources differ by 1,280 parameters. We use the checkpoint-level count here; the discrepancy remains unreconciled.

We note for the reader that 22.8 M of the 39.1 M parameters sit in the *velocity* encoder, not in the wavefield path. Statements of the form “almost all parameters are spectral kernels” are true but misleading for this reason.

The coarse field is learned. One internal branch of the decoder is named `coarse` throughout the code and, for traceability, in this paper. The name is historical and misleading, so we state plainly what it is. It is computed as a low-rank tensor product of four learned branch encodings for velocity, source, travel-time and trunk coordinate features. These are summed over rank and scaled:

$$\text{coarse} = \frac{\gamma}{\sqrt{R}} \sum_{k=1}^R v_k \cdot s_k \cdot \tau_k \cdot u_k,$$

with γ a learned scalar. It is a *neural low-rank pressure field*, evaluated at arbitrary query coordinates rather than on a grid. It is **not** a coarse-grid numerical solution, and the model contains no time-marching loop, no Laplacian and no finite-difference stencil of any kind. The field is queried directly in temporal blocks, without autoregressive feedback.

Both terms of $y = \text{coarse} + s r$ are learned. Their decomposition does not enforce alignment between the correction and the error of the baseline field. A numerical baseline would instead define a specific solver discrepancy as the correction target, although learning that discrepancy would still require validation. Our observations concern this learned two-branch implementation and do not evaluate a solver-in-the-loop method.

Evaluation lineages. The anchor time-domain panels contain frozen training records: 32 uniform, 32 layered, and 96 Marmousi. The 480-record validation spectrum comes from A4 radial-44, checkpoint step 22814, with 16-frame supervision. These endpoints have different histories and are not pooled into a single accuracy result. Validation is used diagnostically; `test_id` remains sealed. Translated Marmousi twins further limit the interpretation of held-out records (§9).

2.3 Metric

Per-record relative L^2 is $\sqrt{N/E}$ with $N = \sum (p - y)^2$ and $E = \sum y^2$ accumulated in float64 over all grid nodes and all frames of the window; it returns undefined when $E = 0$, and an undefined value blocks an aggregate rather than being dropped. The future window is the frames after the 8-frame initial condition; it is split into three equal bands B1/B2/B3 by array partition. Three aggregation conventions are computed: record-equal mean, family-equal mean, and energy-pooled $\sqrt{\sum N / \sum E}$. The 0.1 threshold used in the diagnostics is distinct from the project objective of 0.05 error with $10\times$ acceleration.

Aggregation sensitivity. The same 480-record validation evaluation, same checkpoint, same code, reported under two conventions, moves a single family’s score by -10.9% to $+32.0\%$ (uniform 0.13088 record-equal against 0.17276 pooled; marmousi 0.56535 against 0.50382). It does *not* move the family ordering, which is uniform $\ll$ layered $<$ marmousi under every convention measured. Every table below therefore names its convention and we make no cross-convention comparison.

Late-window denominators. The late-band relative L^2 has a near-zero denominator for the homogeneous family: uniform B3 exceeds 1.0 on 11 of 32 records and its family mean is 1.7456. This is a denominator artefact, and it is never gated and never headlined. True energy pooling removes it entirely ($1.7456 \rightarrow 0.0982$), which is an independent demonstration that it is not a physical result. Figure 2 shows the same artefact as a waveform: on the family-median uniform record the late-band trace relative L^2 is 156.6 while the reference coda it divides by is 1.045×10^5 times smaller than that record’s own first arrival. Marmousi B3 retains substantial signal energy: its mean per-record energy share is 30.6% in the anchor panel. This is why the diagnosis is anchored on Marmousi.

Table 2: Runtime and accuracy, single instance on the same GPU, family-median records, train split. Accuracy is relative L^2 over the future window against the stored truth. ^a includes velocity-resampling discrepancy (see text). ^b 16-frame decoding has matching output peaks, but full-field equivalence and accuracy were not independently checked.

Configuration	Grid	Steps/passes	Runtime (s)	uniform	layered	marmousi
LWC-84, generation protocol	401 ²	8000	20.07	0.000019	0.0176 ^a	0.0581 ^a
LWC-84, native stored grid	201 ²	4000	10.00	0.0177	0.0156	0.1197
Operator 29359, 1-frame decode	—	401	3.97	0.0515	0.2135	0.4009
Operator 29359, 16-frame decode ^b	—	26	2.54	not re-evaluated		

2.4 Runtime against the classical solver

Table 2 compares one operator query against the classical LWC-84 solve on the deployment GPU (RTX 4090D), under one timed protocol: a single instance, the full 401-frame 201×201 pressure field materialised in physical units, CUDA-synchronised timing, no disk I/O inside the timed region. The records are the three family medians that Figures 1 and 2 show, so the accuracy column is the same quantity the rest of the paper reports; runtimes are means over repeated runs. The standard operator time varies by less than 1% and solver times by approximately 2.5% across the timed repeats. Two tested classical configurations define the comparison: a rerun using the generation grid and timestep (401×401 at $\Delta x = 5$ m, $\Delta t = 1.25 \times 10^{-4}$ s, 8000 steps, then restriction to the stored grid), and the native-grid classical configuration that emits the stored product directly (201×201 at $\Delta x = 10$ m, $\Delta t = 2.5 \times 10^{-4}$ s, 4000 steps, no restriction).

The standard decode is $2.5\times$ faster than the tested native-grid solve and $5.1\times$ faster than the fine-grid rerun. Batching 16 query frames reduces operator time to 2.54 s, a 36.0% reduction, giving timing ratios of $3.9\times$ and $7.9\times$, respectively. These are timing ratios, not accuracy-matched speedups: the cheapest tested classical configuration, the native-grid solve at 10.00 s, is *more* accurate than the operator on every family median in Table 2 (0.0177/0.0156/0.1197 against 0.0515/0.2135/0.4009), and every speed statement in this paper carries that caveat. The batched artifact stores matching output peaks, not a full-field equality test; its accuracy is therefore left unreported. Standard decoding spends 3.963 of 3.969 s in the dense decoder, with approximately 6 ms for input preparation.

The output products are matched, but the information supplied to each method differs. The operator receives eight stored initial-condition frames; the solver starts from the prescribed source and medium. Timings exclude model loading, disk I/O, and production of those initial frames. These are conditional query costs, rather than end-to-end deployment costs or an accuracy-matched comparison. There are three figure records, four timed operator repeats per record, and two solver repeats per record after warm-up.

On the Marmousi median record, the standard operator error is $3.3\times$ the native-grid solver error and $6.9\times$ the fine-grid rerun error. The fine rerun uses stored 10 m velocity bilinearly upsampled to 5 m, so its discrepancy includes medium resampling. The uniform rerun, where this resampling is exact, reaches 1.9×10^{-5} relative L^2 . This supports a resampling contribution in heterogeneous media but does not isolate every error source. Operator accuracy here is measured on training records and cannot establish held-out deployment performance.

As an approximate cost account, 4003 solves at the measured fine-grid runtime would require 22.3 GPU-hours. The standard operator saves 16.1 s per query against that configuration, so about 5000 queries would repay this data-generation estimate alone. Training, tuning, and initial-condition acquisition add further cost. An independent validation-record timing run measured 22.1 s mean solver time rather than 20.1 s, indicating session variability. Four timing artifacts and three benchmark scripts are provided in `results/runtime_20260922/`.

Table 3: Anchor 29359, record-equal convention, train split, frozen record lists. † ill-conditioned, reported for completeness only.

Family	n	Full future	B1 (early)	B2 (middle)	B3 (late)
uniform	32	0.060902	0.054989	0.064430	1.745590 [†]
layered	32	0.230264	0.123030	0.398474	0.760080
marmousi	96	0.396499	0.175279	0.359535	0.572774

Table 4: Energy share against error-mass share, anchor 29359, record-equal, train split.

Family	Energy share B1/B2/B3	Error-mass share B1/B2/B3
uniform ($n = 32$)	0.6793/0.2646/0.0561	0.6055 /0.2892/0.1053
layered ($n = 32$)	0.7734/0.1882/0.0384	0.2503/ 0.4982 /0.2515
marmousi ($n = 96$)	0.3372/0.3567/0.3061	0.0672/0.2918/ 0.6410

3 The error floor

For layered and Marmousi media, the mean error increases across the three time thirds. Marmousi B3 is $3.27\times$ its own B1; the uniform late score is denominator-sensitive. Not one of 96 Marmousi records reaches 0.1 on the full window, nor in B2 or B3; the within-family spread is $p_{90}/p_{10} = 2.02$.

Figure 1 shows what that growth looks like in space, for the record selected by a rule fixed in advance: the lower median of the family’s 96 frozen records by full-future relative L^2 . The prediction tracks the direct wavefront throughout the window. What it loses is the scattered coda behind that wavefront: the difference field grows from 0.292 of the reference peak at the middle of B1 to 0.544 in B2 and 0.797 in B3, and by the late band it is spread over the whole domain rather than sitting on any one interface. The per-frame curve in the same figure falls to 0.1060 at $t = 0.115$ s, once the initial transient has passed, and then climbs to 0.6540 at the final frame, which is its maximum over the window. The band averages of Table 3 are therefore not an artefact of where the two band boundaries happen to fall.

3.1 Error mass versus energy

For a record with band truth energies E_b and per-band relative L^2 values r_b , write the band energy share $s_b = E_b / \sum_j E_j$. The identity

$$\text{full}^2 = \sum_b s_b r_b^2 \quad (1)$$

is exact, and the *error-mass share* is $m_b = s_b r_b^2 / \sum_j s_j r_j^2$. Evaluated over all 160 anchor records, the maximum absolute residual of (1) is 1.055×10^{-15} (maximum relative residual 6.14×10^{-15}). The identity holds at machine precision.

Three readings follow. Marmousi energy is nearly flat in time, so the late window is not a tail of vanishing signal, yet 64.1% of its error mass sits there. The families fail in different places: uniform fails *early* (60.6% of its error mass), layered in the middle. Any intervention that trades early accuracy for late accuracy therefore damages the family whose training-panel mean is below the 0.1 diagnostic threshold.

An intervention confined to the early and middle windows cannot remove the observed late error. If those windows were exact and B3 remained unchanged, each record would retain

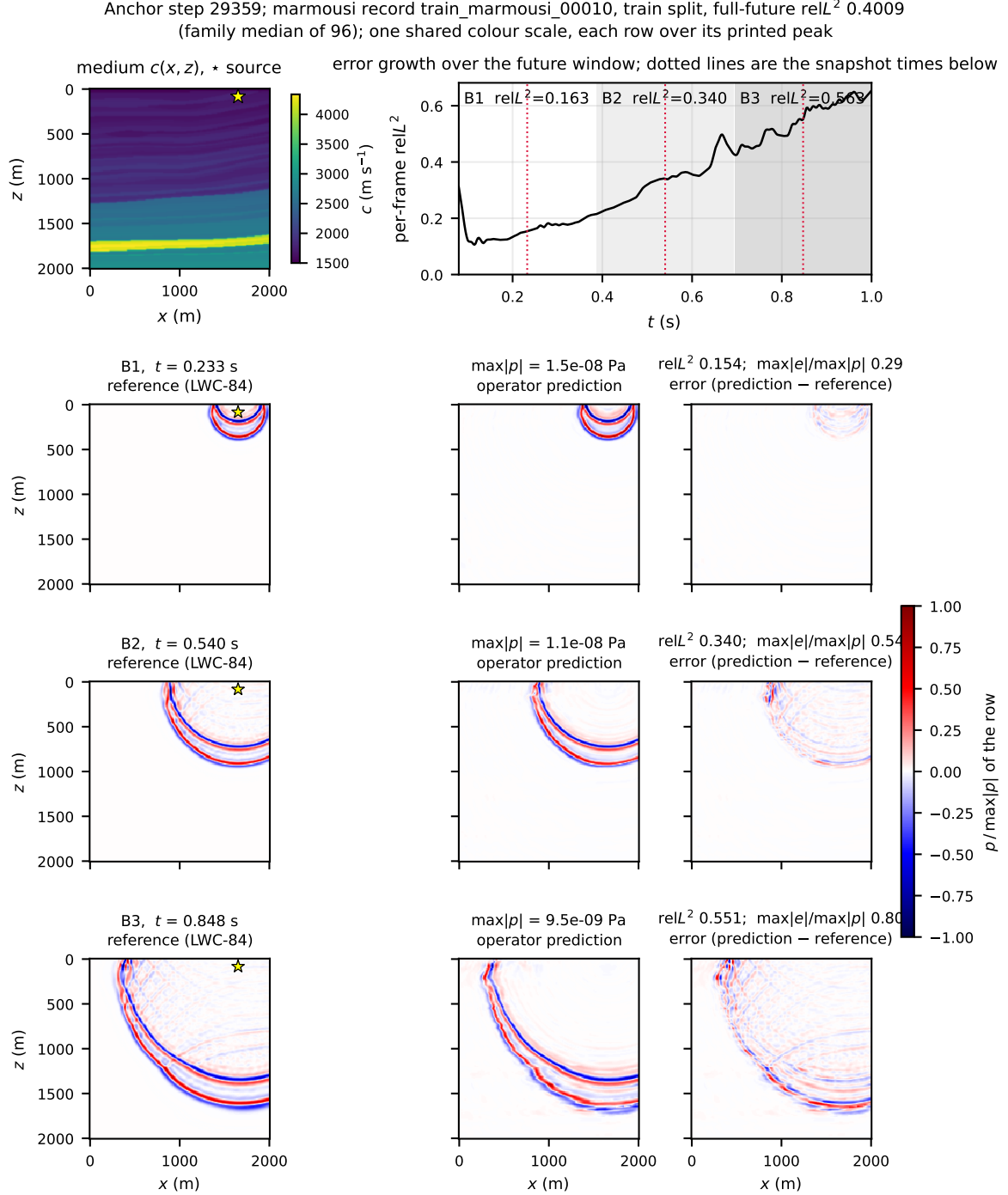


Figure 1: Wavefield snapshots at the anchor, for the family-median Marmousi record (`train_marmousi_00010`, full-future relative L^2 0.4009 against the family mean 0.3965). Top: the velocity model with the source starred, and the per-frame relative L^2 over the future window, shaded by the three time bands, with the snapshot times marked. Rows: reference LWC-84 field, operator prediction, and their difference at each band’s mid-frame. All nine panels share one dimensionless colour scale; each row is divided by its own peak, printed on the row in Pa, so the error panel reads directly as the fraction of the local signal the operator gets wrong ($0.29 \rightarrow 0.54 \rightarrow 0.80$). Fields are displayed $[z, x]$ exactly as stored. Train split, checkpoint step 29359; the plotted scores are bitwise identical to the panel rows behind Table 3, because the figure imports the same measurement module at the same `sha256` rather than reimplementing the metric.

relative error $\sqrt{s_{i,B3}} r_{i,B3}$. The corresponding aggregate quantities are

$$r_{\text{late,record}} = \frac{1}{n} \sum_i \sqrt{s_{i,B3}} r_{i,B3}, \quad r_{\text{late,pooled}} = \sqrt{\frac{\sum_i N_{i,B3}}{\sum_i E_i}}. \quad (2)$$

Recomputation from the stored per-record scalars gives Marmousi counterfactuals of 0.312137 record-equal and 0.287059 pooled, respectively $3.12\times$ and $2.87\times$ the target. The arithmetic and source scalars are included in `results/diagnostic_audit.20260922/`. Multiplying the family-average energy share and family-average relative error would give 0.3169, but that operation is not an exact record-equal aggregate and is not used as a bound. This calculation motivates reducing late error directly. It does not exclude a curriculum or weighting change that actually improves late predictions.

Figure 2 makes the same point on a single receiver trace and adds the cross-family contrast. Over the full future window all three predictions reproduce the first arrival, and the uniform and layered traces are visually indistinguishable from the reference there. The late band has to be re-normalised before it is visible at all, because the coda sits a factor of 12.4 (Marmousi) to 88.8 (layered) below the first-arrival peak; once it is, the Marmousi prediction is wrong in both amplitude and phase while the reference is still ringing, and its B3 trace relative L^2 is 1.1123 against 0.5186 over the full window. This is the same concentration Table 4 reports as an energy-weighted average, seen at one node.

The uniform panel of that figure is the near-zero-denominator case, and the figure labels it as such. Its coda has left the domain through the CPML, so the re-normalisation gain is 1.045×10^5 and the B3 trace relative L^2 is 156.6 — a division artefact of exactly the kind §2.3 excludes from every claim in this paper, shown here so that the exclusion can be checked rather than taken on trust.

4 Comparison with classical solvers and an operator baseline

This section adds three controlled comparisons: coarse-grid classical solves that exhibit textbook numerical dispersion, receiver-level and snapshot-level views of how the neural error differs from that dispersion, and a preregistered equal-budget DeepONet baseline. All wave-field figures and per-record numbers in this section come from *train-development monitoring records* (the family-median records of a 9-record dev set, 3 per family); they illustrate error character and are *not* held-out certification. The evaluated network is a subsequent lineage (`scno_homogeneous_longrun_gap15_v1`, update 5000, 39,140,293 parameters, trained on a train-only frequency-gap fork of the same generation pipeline), not the anchor of §3; the anchor appears in this section only in the runtime rows, where it is labelled. On the two figure records the future-window relative L^2 is 0.2785 (`train_marmousi_00076`) and 0.1960 (`train_layered_00299`), verified by recomputation against the stored evaluation to 10^{-6} .

4.1 Coarse-grid classical dispersion versus the neural error

The dataset’s own LWC-84 + CFS-CPML solver was re-run on coarse grids for the two figure records, interpolated back to the stored 201×201 grid (bilinear, exact at shared nodes), and scored with the paper metric. The solver requires odd node-centred grids and raises on even n_x/n_z , so the nearest admissible configuration to a 50×50 grid is 51×51 at $\Delta x = 40$ m (coarse nodes exactly on fine nodes, stride 4); 101×101 at $\Delta x = 20$ m is a second point on the resolution trend. Velocities are nodally decimated without anti-alias smoothing, so the coarse totals are *coarse-grid solution error* (time-stepping dispersion plus medium-sampling error), both unavoidable at that resolution. Time stepping, source injection, and boundary treatment follow the frozen native-grid benchmark protocol (`results/paper_comparisons_20260923/`, group 3).

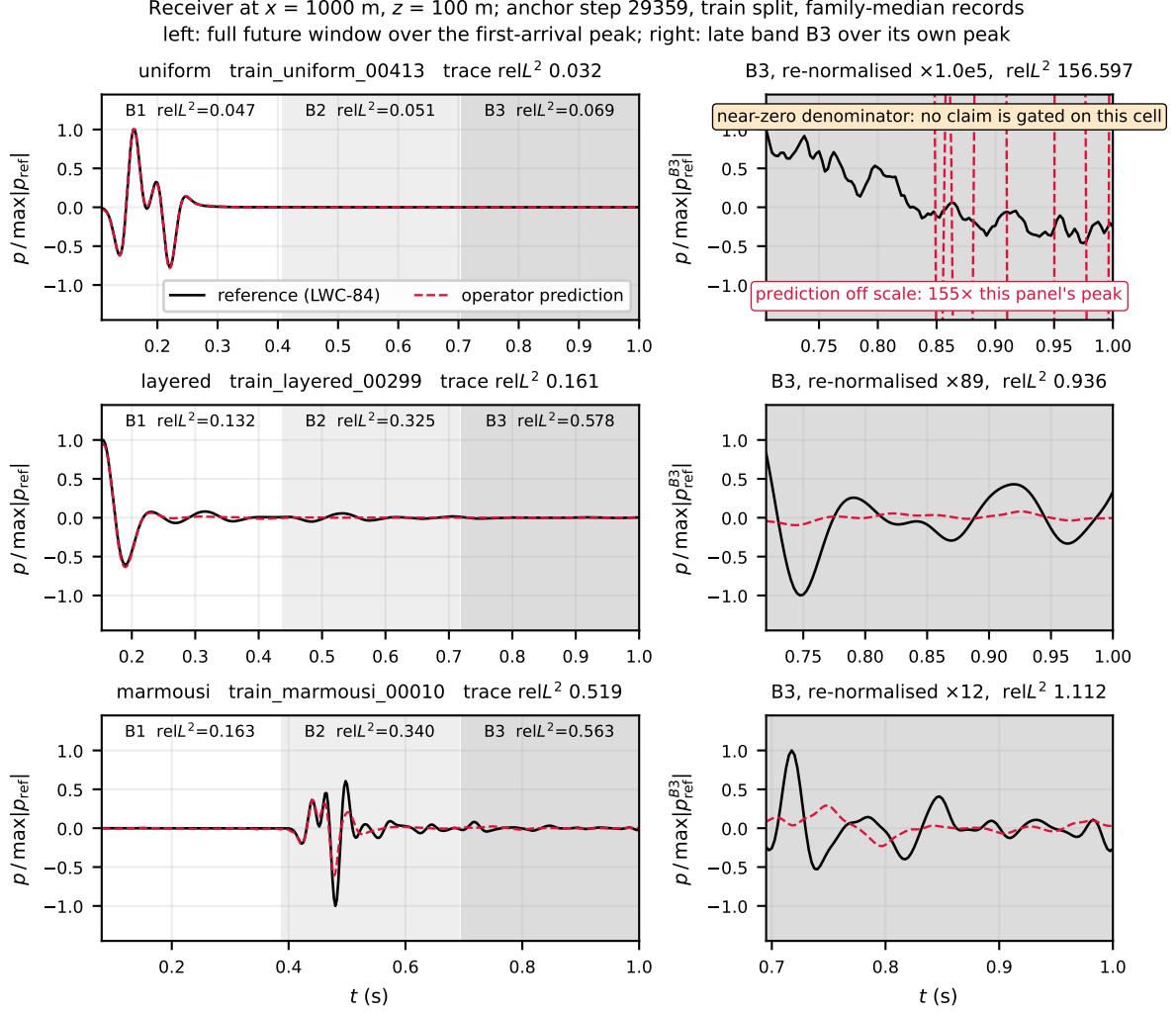


Figure 2: Receiver waveforms at the anchor, at $x = 1000$ m and $z = 100$ m, for the family-median record of each family. The receiver line sits 100 m below the surface because the free surface is node-centred with $p(z = 0) = 0$ identically. Each window starts at that record's source onset plus its 8-frame initial condition, so the three time axes begin at different instants. Left: normalised by the reference first-arrival peak, shaded by the three time bands with each band's field-wide relative L^2 printed. Right: the late band alone, re-normalised by its own reference peak, with the gain that re-normalisation applies stated on the panel. The uniform late panel is flagged because its denominator is near machine zero; no claim in this paper is gated on that cell, and its prediction leaves the re-normalised window entirely. Train split, checkpoint step 29359.

Table 5: Coarse-grid classical solves against the neural operator, future-window relative L^2 (bands B1/B2/B3 in parentheses), train-development records. The coarse columns include medium-sampling error (see text). At $\Delta x = 40$ m the Marmousi record has ≈ 3.3 points per minimum wavelength ($v_{\min} = 1500$ m/s), below the eighth-order comfort zone; the layered record retains ≈ 4.5 ($v_{\min} = 2068$ m/s).

Record	Neural operator	LWC-84 51×51 ($\Delta x = 40$ m)	LWC-84 101×101 ($\Delta x = 20$ m)
marmousi_00076	0.279 (0.124/0.245/0.409)	0.514 (0.470/0.514/0.558)	0.120 (0.122/0.118/0.119)
layered_00299	0.196 (0.108/0.311/0.552)	0.103 (0.066/0.138/0.328)	0.030 (0.020/0.046/0.065)

Table 5 and Figure 3 show two distinct error mechanisms. The 51×51 solve on the Marmousi record carries the classical dispersion signature: receiver arrivals late by $+2.5$ to $+7.5$ ms at three of four receivers, the receiver spectral centroid downshifted from 12.97 to 10.66 Hz, and a per-frame error that is roughly flat in time (bands 0.470/0.514/0.558). The neural operator error has the opposite structure: arrivals in phase (lag 0 ms at most receivers; centroid 13.07 Hz), but error *growing* in time (0.124 $\rightarrow$ 0.409 across bands) as a late-coda amplitude deficit (receiver amplitude ratio 0.60 at $x = 400$ m on the Marmousi record).

This dispersion argument is supported only by the Marmousi record. On the layered record the 51×51 solve is *more* accurate than the neural operator (0.103 against 0.196), because $v_{\min} = 2068$ m/s keeps ≈ 4.5 points per wavelength at $\Delta x = 40$ m and its receiver lags are 0 ms; the 101×101 solve is more accurate than the operator on both records (0.120 and 0.030). We report both families; the neural operator does not dominate coarse classical solves, and the contrast in Figure 3 is an error-character contrast, not an accuracy ranking.

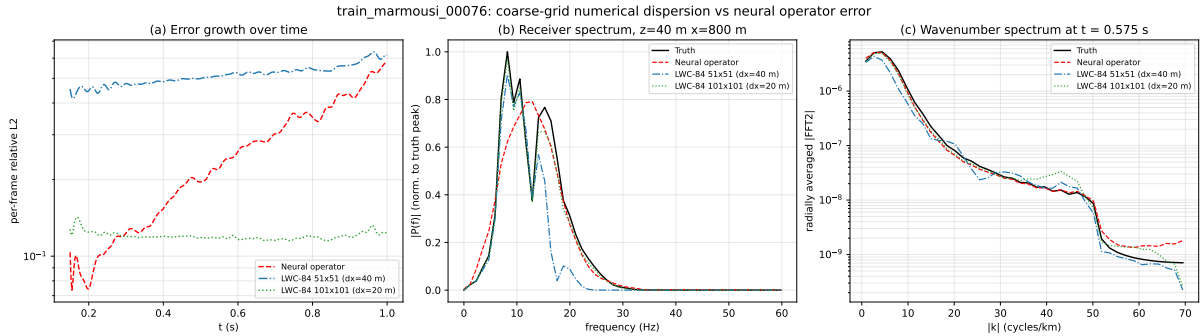


Figure 3: Coarse-grid numerical dispersion against neural-operator error on the train-development Marmousi record `train_marmousi_00076` (not a held-out certification). Left: per-frame relative L^2 over the future window. Middle: receiver amplitude spectrum at $z = 40$ m, $x = 800$ m; the 51×51 solve loses the band above ≈ 15 Hz (centroid 12.97 $\rightarrow$ 10.66 Hz) while the neural trace keeps the truth band. Right: radially averaged wavenumber spectrum at $t = 0.575$ s. Fields are $[z, x]$ as stored.

4.2 Snapshots and receiver waveforms

Figures 4 and 5 show truth, prediction, and difference at early/middle/late physical times for the two records. Both records reproduce the direct wavefront and lose scattered coda with time: per-frame relative L^2 grows from 0.094 ($t = 0.2175$ s) to 0.242 ($t = 0.575$ s) to 0.580 ($t = 1.0$ s) on the Marmousi record, and from 0.045 to 0.318 to 1.010 on the layered record, whose final stored frame is at the trivial-predictor level while its full-window score remains 0.1960. This is the same late-time structure as the anchor diagnosis of §3, on a different lineage.

Figure 6 overlays the 51×51 coarse solve on the receiver waveforms. The coarse trace arrives

late and rings (dispersive tail); the neural trace stays in phase through the first arrivals and then loses late-coda amplitude. The receiver nodes at $x \in \{400, 800, 1200, 1600\}$ m, $z = 40$ m are native coarse nodes (stride-4 multiples), so the plotted coarse traces are solver values, not interpolants.

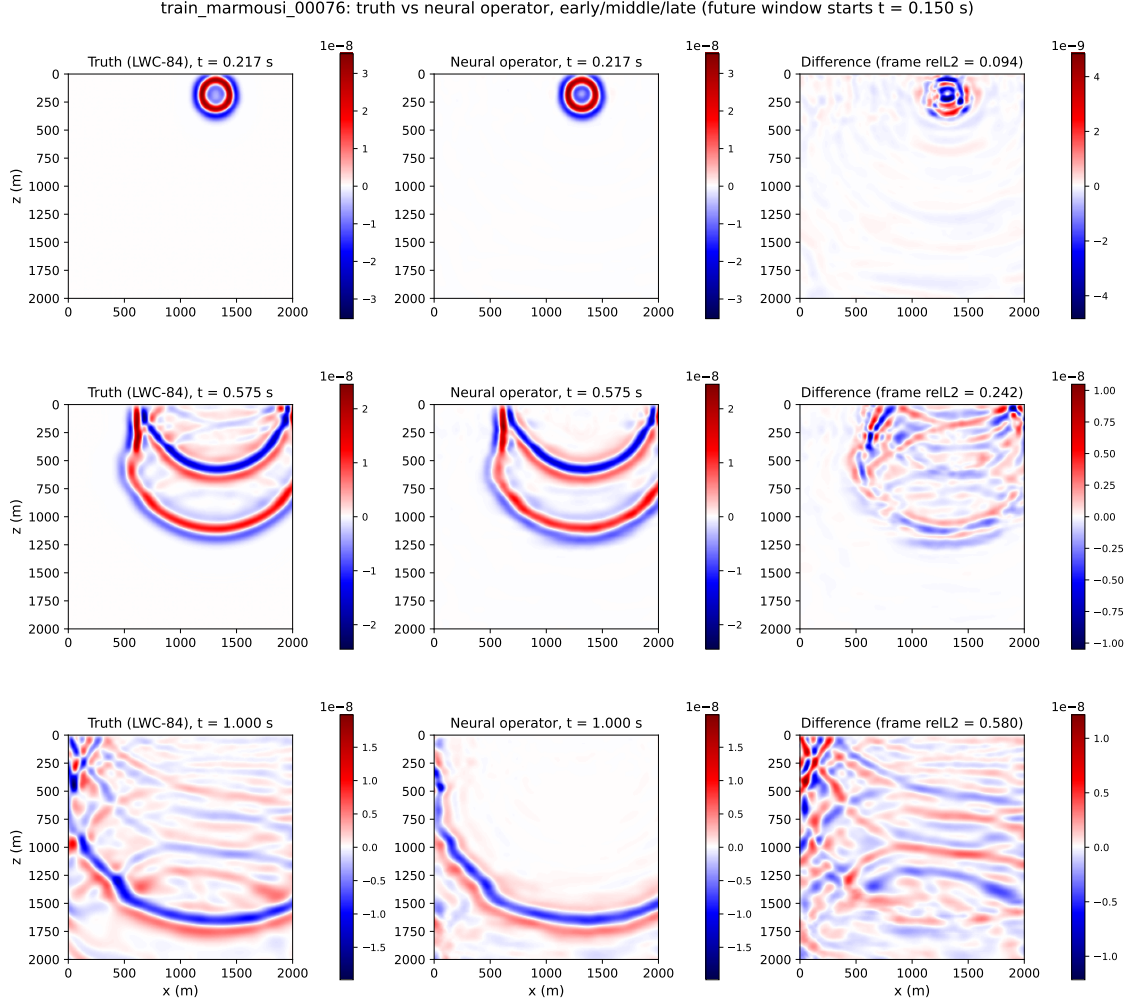


Figure 4: Truth, neural prediction, and difference at early/middle/late times on the train-development Marmousi record `train_marmousi_00076` (not held-out certification). Truth and prediction share one symmetric colour scale per row; the difference column has its own symmetric scale per row. Per-frame relative L^2 is printed on each difference panel. Fields are displayed $[z, x]$ exactly as stored, depth increasing downward.

4.3 Runtime in context

The timing protocol and numbers are those of §2.4 (Table 2); no new timing runs were performed. Assembled record-matched ratios: the anchor operator is $5.06\times$ (standard) and $7.90\times$ (16-frame batched) faster than the fine-grid protocol, and $2.52\times/3.93\times$ faster than the native 201×201 solve; the independent generation-protocol timing on validation records is 22.06 s. Two caveats bound any speed claim. First, the cheapest tested classical configuration (native grid, 10.00 s) is more accurate than the anchor operator on every family-median record in Table 2, so the speed advantage carries an accuracy cost on every family. Second, training is amortised and excluded; no end-to-end (training-inclusive) speedup is claimed.

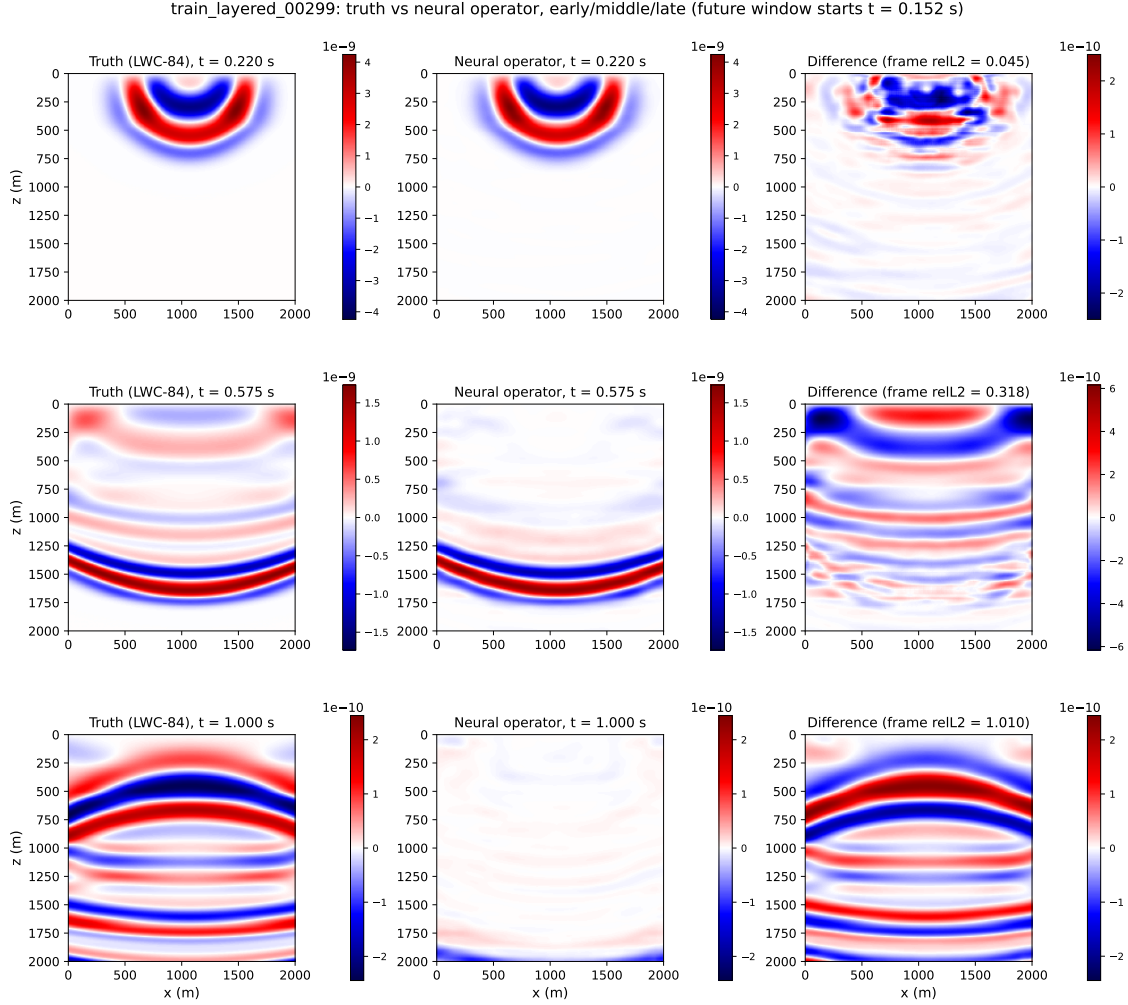


Figure 5: As Figure 4, for `train_layered_00299` (train-development record; not held-out certification). The final stored frame reaches per-frame relative L^2 1.010: the prediction retains almost none of the reverberating coda at $t = 1.0$ s, although the full-window score is 0.1960.

4.4 An equal-budget DeepONet baseline failed to train

A preregistered DeepONet baseline at matched parameter budget (39,975,985 against our 40,484,837; budget $\pm 2\%$; the reference is the IC8 40-M configuration behind the A4 lineage, not the 39.1-M anchor of §2) was trained on the same task. Its main run was terminated by SIGTERM at step 4,862 of a planned 22,814, so its numbers are not a converged endpoint. At every recorded validation on its fixed 12-record checkpoint-selection batch, its dense relative L^2 was 1.000012–1.000014, indistinguishable from the trivial zero predictor on all three families. A subsequent family-level probe (12 train records, 2000 steps) found the control arm never below 1.0015, and an anti-collapse arm oscillating at 0.97–1.33, not separable from control on any of its three effect axes; single-record fits do converge (relative L^2 0.025 at 2000 steps), so the recorded verdict attributes the failure to optimisation, not capacity.

Table 6 states the recorded facts. Because the baseline collapsed, these numbers support only the statement that *this DeepONet configuration failed to train in this setting under its preregistered protocol*; they do not support a claim that our operator outperforms a converged DeepONet, and we make no such claim.

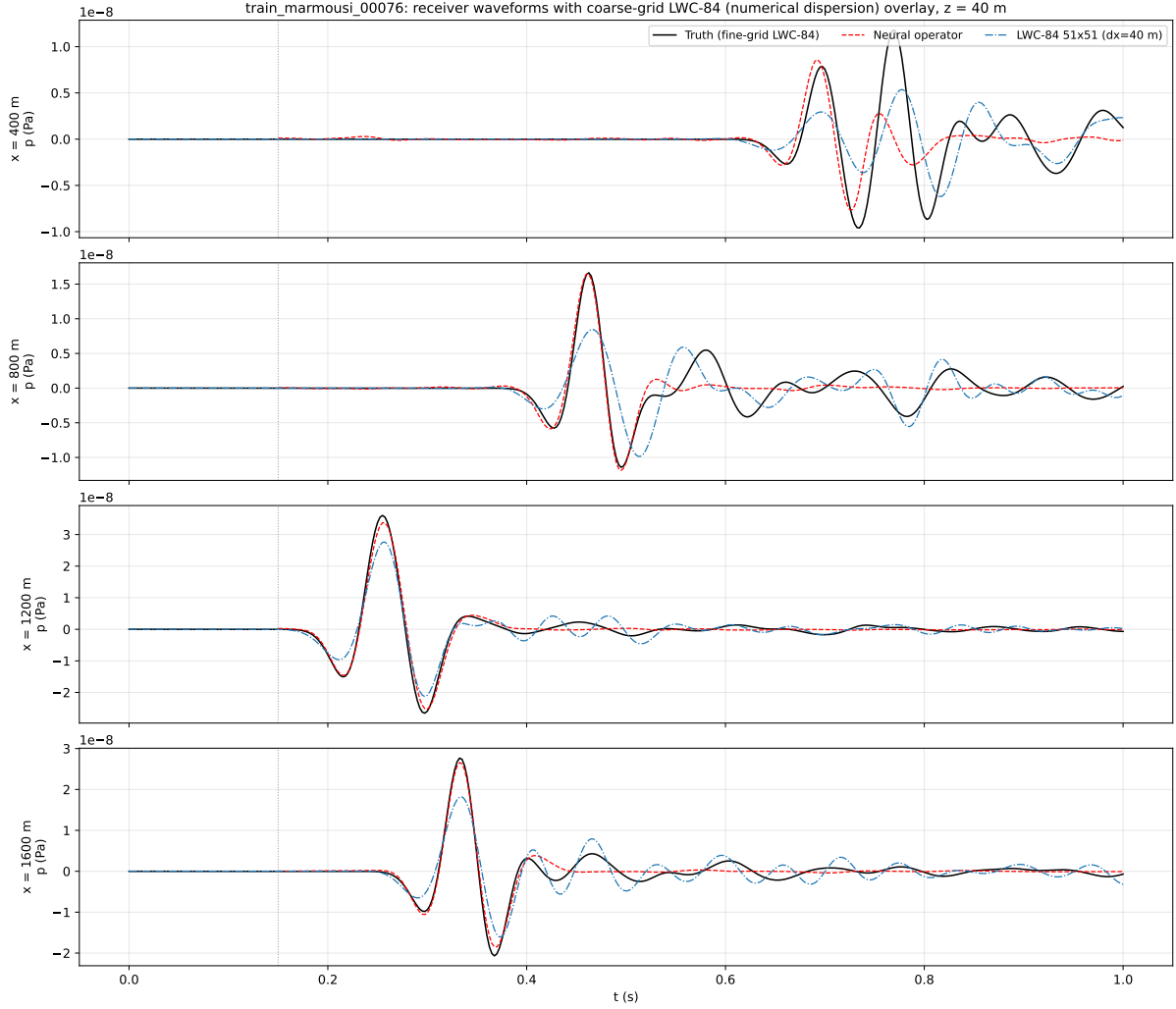


Figure 6: Receiver waveforms at $z = 40$ m for `train_marmousi_00076` (train-development record; not held-out certification): truth, neural operator, and the 51×51 coarse LWC-84 solve. The coarse trace shows phase lag and ringing (numerical dispersion); the neural trace is in phase but under-predicts the late coda. The dotted line marks the start of the future window.

Table 6: DeepONet baseline at matched parameter budget. No same-caliber comparison exists: the DeepONet run stopped early and was never evaluated on the 480-record census, so the rows pair the nearest available calibers and each row names its own. The comparison establishes that this baseline failed to train in this setting; it is not an accuracy ranking between converged models.

Quantity	DeepONet 40M	Ours (ICS 40M)	Caliber
Parameters	39,975,985	40,484,837	same $\pm 2\%$ budget (preregistered)
Fixed 12-record validation, dense rel. L^2 (best, step 4488)	1.000012	n/a (not our selection metric)	DeepONet's selection batch; ≈ 1.0 = trivial
480-record validation census, dense future rel. L^2	not evaluated (stopped at step 4862/22814)	0.352 (0.131/0.302/0.565)	ours: full census, A4 step 22814
12-record family probe (2000 steps), point rel. L^2	control 1.0015–1.1168, never < 1.0	n/a	packaged-loss probe caliber

5 Intervention tests and their scope

Table 7 summarizes the tested alternatives. Frozen protocols accompany the major intervention arms. Each result constrains the specified implementation, data panel, and budget; it does not exclude the broader method class.

Table 7: Diagnostic tests and their scope. Negative findings apply to the tested settings.

Candidate	Measurement and interpretation
Spectral truncation band-width	Truth energy outside the anchor’s radial-44 kernel is 0.882%; zeroing <i>all</i> out-of-band error buys at most $1.02\times$ ($1.27\times$ at rect16) against $4.05\times$ needed.
The error is high-wavenumber	Band-3 truth is <i>narrower</i> than band-1 (10.85% against 14.71% outside rect16) while its error is $2.4\times$ larger.
Width / capacity	An internal width sweep saturates near 0.125; this does not exclude different capacity allocation or architectures.
Optimiser / gradient clipping	Releasing the clip is +135.8% (clip = 20) and +163.8% (clip = 10^9), worse on 44/44 paired steps. Steps are serially dependent, so this is a descriptive comparison.
Inference-time iteration of the trained correction	$n = 2$ is +1.21% (19/20 worse); $n = 3$ is +3.69% (20/20 worse).
One proposed reflection carrier	The strongest-impedance single bounce arrives at 1.62–1.75s, outside the 1.0s record. This excludes that path as an in-window carrier, not other scattering paths.
A bad tail of records	Forcing the worst 30% to 0.1 reaches only 0.2698; putting <i>every</i> record at the current p_{10} reaches only 0.2657.
Sampling energy-floor artefact	The floor fires on 0.0% of Marmousi frames (uniform 23.43%, layered 2.02%).
Frame re-weighting	The weighted mean is invariant to $w \rightarrow cw$, so “down-weight early” and “up-weight late” are equivalent only when their normalized relative weights are identical; general re-weighting remains open.

The deployed discrete residual changes the supervised norm. The source-aware LWC-84 increment term was enabled at weight 0.01 in 37 production runs, including the anchor lineage. Its weighted objective share is 0.0038%–0.18%. This is a small value contribution, although loss magnitude alone does not determine its gradient contribution.

For the deployed defect-matched form, the linear increment operator A gives $A(\text{pred}) - A(\text{label}) = A(\text{pred} - \text{label})$. The audit confirms this identity over 288 triplets, with maximum relative deviation 4.57×10^{-13} . The term therefore provides a physics-derived weighting of supervised error, rather than an independent equation constraint at unlabeled times. Such weighting may still improve optimization or accuracy; the identity does not prove that increasing its weight cannot help.

Its weighting differs substantially from the field-error distribution. In a six-band audit, 79.9% of error energy lies in the smoothest band, compared with 27.3% of $\|Ae\|^2$. The penalty amplifies high-wavenumber errors, and its framewise rank correlation with relative L^2 changes sign across families. These observations motivate gradient and transfer checks before changing its weight. They do not establish the outcome of an unrun weight sweep.

A label-free residual needs a discretization consistent with the stored product. Labels are lowpass restrictions of a finer solve, and their own late Marmousi defect is 1.069% under the audited operator. Stored cadence is 2.5ms on 10m spacing, versus 0.125ms on 5m spacing during generation. At 7963m/s these imply Courant numbers of 1.99 and 0.199. A large-step increment residual need not approximate the generator accurately; this is not by itself a statement about neural-network stability. The radius-20 interior margin also excludes roughly 36% of the stored area. Free-surface extensions and CPML memory states require explicit treatment. We therefore leave weak-form, substepped, and boundary-aware residuals as distinct

Table 8: Counter-experiment arm at step 48059 against the anchor, record-equal, train split.

Cell	Anchor	Arm	Δ relative
Marmousi B3 (main gate, $n = 96$)	0.572774	0.588998	+2.83%
Marmousi full ($n = 96$)	0.396499	0.423210	+6.74%
layered full ($n = 32$)	0.230264	0.258124	+12.10%
uniform full ($n = 32$)	0.060902	0.116177	+90.76%

experiments.

5.1 The counter-experiment

To test whether releasing the collapsed correction branch and switching to a phase-free rectangular spectral kernel breaks the floor, an 18700-step arm was run from the anchor to step 48059 under a frozen pre-registration. Table 8 gives the result: the main gate required $B3(M96) \leq 0.5441354$ and measured 0.5889980. Every family got worse. Run integrity was clean: 18700 contiguous steps, zero duplicates, zero rejected retries, zero non-finite events.

Training loss fell by roughly 6% over the window, the only one of four arms to leave its plateau, while error on the frozen training-panel evaluation rose in every family. The per-record paired damage fits the descriptive relationship $b = \sqrt{a^2 + c^2}$. The fitted c values are 0.09846, 0.11119, and 0.14661 for uniform, layered, and Marmousi, respectively. The corresponding R^2 values are -0.460 , 0.9560 , and 0.9417 . Record ranking is largely preserved (Spearman 0.978 and 0.969 on layered and Marmousi). This fit describes the paired scores for layered and Marmousi records. It is consistent with an added error component, but scalar error norms do not establish vector orthogonality or identify its cause. Several settings changed together, so the arm cannot isolate kernel shape from gating or supervision effects. The uniform fit is poor ($R^2 < 0$).

6 Temporal-transfer and correction diagnostics

6.1 Limited transfer between supervised and unsupervised frames

In a single-record adaptation probe with 16 supervised frames, relative L^2 on a disjoint 16-frame late set changes from 0.37067 to 0.36479, a ratio of 0.9841. The temporal offsets are only 2.5–17.5 ms. Increasing supervision to 51 frames gives 0.19163, a ratio of 0.5170 to the same initial error. It also increases the number of evaluation frames within 10 ms of a supervised frame from 0 to 11 out of 16.

The deployed supervision spacing is a mean gap of 56.78 ms with *no* gap below 10 ms, while the truth’s own correlation drops below 0.5 between 10 and 20 ms. The probe therefore requires interpolation across gaps wider than the measured decorrelation scale. Its limited transfer is consistent with frame-specific adaptation. The 51-frame probe demonstrates a within-record density effect, but not cross-medium generalization or a validation-wide accuracy improvement.

6.2 Scale collapse of the correction branch

The decoder forms $y_d = c + s r$, followed by the fixed free-surface factor $F(z) = \tanh^2(z/(20 \text{ m}))$. The deployed prediction is therefore $F(c + s r)$. Define the deployed baseline error $e = Fc - y_{\text{truth}}$ and correction $q = Fr$. For fixed nonzero e and q , least squares gives

$$\min_g \frac{\|e + gq\|}{\|e\|} = \sqrt{1 - \cos^2(q, -e)}, \quad g^* = \frac{\langle -e, q \rangle}{\|q\|^2}. \quad (3)$$

This identity bounds rescaling of a fixed correction direction. It does not bound a branch whose features are retrained.

A forward-only audit captures the pre-gate correction directly and evaluates 144 frames from 12 training records. The median optimal ratio in deployed output space is 0.999234, a 0.0766% median reduction relative to the coarse-only error. This oracle fits an independent scalar to each frame using its true residual. It is consequently more permissive than a single deployable gain, and is not a family-level accuracy score. The recorded reconstruction check has maximum relative error 1.03×10^{-15} , with weights unchanged bitwise.

An earlier probe omitted F from the compared decoder tensors and reported 0.9984. That decoder-space number is superseded for deployment interpretation. The corrected audit and its script are included in `results/diagnostic_audit_20260922/`. The anchor gate remains $s = 1.0425 \times 10^{-4}$; neither its size nor the oracle proves that training found a globally optimal decomposition. The result shows limited improvement from rescaling the current direction and motivates learning a different direction.

Gate-dependent gradients motivate a different training target. A frozen-output scale sweep measures log-log gradient slopes near one for several correction modules: 1.041 for the output weight and 1.007 for the dense spectral kernels. Parameters outside the dense decoder have a near-zero slope. This agrees with the chain-rule factor s in $\partial L / \partial W_r$. The measured objective increases when the current frozen branch is opened, including +129.8% at $s = 1$. These measurements explain why opening this trained gate alone is unpromising; they do not prove that joint retraining has the same optimum.

Directly supervising the deployed correction $q = Fr$ against the detached target $y_{\text{truth}} - Fc$ would provide a branch gradient without that multiplicative gate. A frozen-feature least-squares probe reports a transferable ratio of 0.9998, so any improvement would require learning new features. Changes to attention, local convolutions, initialization, or staged optimization remain untested. Sharing a scalar gate does not prove that another architecture must learn the same correction direction.

7 Frequency-resolved error and conditional improvement estimates

Using the frequency-resolved error decomposition over the 480-record validation evaluation (four radial wavenumber bands, Parseval verified per frame), band relative L^2 rises monotonically with wavenumber: 0.3641/0.5263/0.7566/0.9924 pooled. Independently, the prediction carries *less* above-mode-16 energy than the truth in all three families (0.1669 against 0.1871 for uniform, 0.0465 against 0.0598 for layered, 0.2100 against 0.2299 for marmousi). Together these indicate an endpoint frequency-dependent error pattern. Learning dynamics would be needed to establish when each frequency is learned.

We use two counterfactuals to quantify the error remaining under restricted changes. One sets bands 3 and 4 to zero error while preserving lower bands. The other gives every band the current lowest-band relative error. Because $\sum_b s_b = 1$, this second score equals the current lowest-band value. Neither counterfactual is a demonstrated intervention or an upper bound on methods that also reduce low-band error. Table 9 reports both.

Most error mass lies in band 2 (64.7% pooled), with only 0.5% in the highest band. Eliminating bands 3 and 4 alone reduces pooled error by 5.1%. Marmousi has the smallest relative enrichment of error in band 3 among the three families: its band-3 error enrichment is 2.309 against 8.690 and 9.467 for uniform and layered, and its truth is the widest-band of the three. Its substantial low- and middle-band errors motivate interventions beyond the spectral tail. This endpoint comparison does not exclude dynamical spectral bias.

Marmousi band 4 has relative L^2 of 1.0412, and uniform band 4 falls below band 3. Both bands contain only about 0.1% of total truth energy and contribute little to the pooled score.

Table 9: Conditional spectral counterfactuals, A4 step 22814, validation split, 480 records.

Family	n	pooled	bands 3+4 $\rightarrow$ 0	fully flattened	gap to 0.1
uniform	90	0.17276	0.1557 (−9.9%)	0.1114 (−35.5%)	$1.11\times$
layered	240	0.28986	0.2827 (−2.5%)	0.2369 (−18.3%)	$2.37\times$
marmousi	150	0.50382	0.4779 (−5.1%)	0.3980 (−21.0%)	$3.98\times$
pooled	480	0.47588	0.4514 (−5.1%)	0.3641 (−23.5%)	$3.64\times$

Small relative energy alone does not establish a numerically ill-conditioned denominator; that interpretation requires absolute-energy and noise-floor checks.

8 Related work

Learned background and correction branches have precedent. Balaji et al. [4] combine an FNO background with a transformer correction for high-contrast Helmholtz scattering. We make no novelty claim for a two-branch decomposition. Their frequency-domain setting, architecture, and training differ from our time-domain full-wavefield task. Their reported success establishes a relevant precedent, but does not isolate attention as the cause of improved accuracy. Our contribution is the diagnostic analysis of one trained implementation and the scope of its failed interventions.

Frequency diagnostics connect to rollout research. [2] identifies the neglect of non-dominant spatial frequency information, often high frequencies, as the primary pitfall limiting stable accurate rollouts. Our measurement that the prediction carries less above-mode-16 energy than the truth in all three families is a related endpoint observation in a single-shot setting, without testing the same rollout mechanism. [3] argues that spectral bias in operator learning is dynamical rather than purely representational and recommends curvature-aware optimisation and spectrally informed losses; both are untested here (§9).

Structured spectral rollout methods do not transfer directly. [6] reports large long-horizon gains for a structured spectral autoregressive operator, but states that it is designed for periodic linear and semilinear evolution PDEs with Fourier multiplier linear dynamics. Our problem is variable-coefficient, with CPML absorbing boundaries and a free surface, and is neither periodic nor a Fourier multiplier. We cite the method for its diagnosis of spectral amplitude and phase drift, and we do not transfer its reported factors to our setting.

Learned dispersion correction: the nearest precedent is fixed-velocity. [7] treats an inaccurate finite-difference Laplacian as incomplete physics and intersperses CNNs between low-fidelity timesteps to limit numerical dispersion during propagation. It is a proof of concept that holds the velocity model fixed and varies only source locations. [5] benchmarks twelve correction architectures for 2-D anisotropic acoustic dispersion over 27,000 heterogeneous velocity fields, finding the FNO best but by a modest margin over a CNN, and finding that pipelines with a classical initial guess substantially outperform those learning the correction directly; it operates on traces rather than full wavefields. Coarse-grid-plus-learning accounting in other domains closes favourably. [8] integrates 1-D equations at 4–8 $\times$ coarser resolution and [9] matches baseline solvers at 8–10 $\times$ finer resolution for 2-D turbulence, a 40–80 $\times$ net speedup with stability over long simulations. The former is 1-D and the latter is Navier–Stokes, a first-order-in-time advection-dominated system, so neither transfers to a second-order hyperbolic problem without demonstration.

9 Limitations

Split leakage is disclosed. The Marmousi family contains 50 m-stride translated twins, with 30 of 30 audited models having a twin. Marmousi numbers on held-out records are therefore *not* clean generalisation measurements.

Lineage. The band decomposition is the validation split at a radial-44 endpoint with 16-frame supervision. The counter-experiment arm is rectangular-kernel with 16 frames. The subsequent density arm uses 51 frames and is a separate experiment. The dense spectral kernel sets the wavenumber response, so the band table does not characterise the rectangular-kernel lineage, and no frequency-resolved product exists for that lineage on either split.

Learning dynamics are not measured. Whether high-wavenumber bands converge later than low ones is not answered here. This distinction separates learning dynamics from endpoint representational error. The frequency-resolved metric is driven by offline evaluation, which was run at a single endpoint; answering it requires evaluation at intermediate checkpoints.

Temporal coverage remains an open intervention. The source-relative sampling window contains 318–372 candidate frames per record. A 51-frame coverage audit reports a fraction 0.736 unvisited under the deployed deterministic placement at both 16 and 64 simulated visits. Alternative full-support and residual-adaptive policies leave fractions 0.577 and 0.149 unvisited at 16 visits, respectively. The latter falls to 0.075 at 64 visits. For Marmousi B3, the reported unvisited fractions are 82.9% and 10.8% for deployed and residual-adaptive placement. The simulation disables continuous-time query flags, whereas the production configuration uses a continuous fraction of 0.125. These are coverage measurements for that simulated schedule, not accuracy gains or a direct audit of the anchor’s 16-frame production schedule. An unvisited frame under finitely many random draws is distinct from a frame with zero sampling probability.

A longer density arm was interrupted before an endpoint gate. A 150-step sampling comparison is insufficient to establish an accuracy improvement, and its arms supervise different frames. Future comparisons should use a fixed evaluation grid selected independently of both samplers, matched compute, and repeated seeds. Every candidate frame should retain a nonzero sampling probability. The same-protocol validation gate must pass before claiming improved long-time accuracy.

Time-conditioned spectral responses are untested. Our dense decoder applies one shared set of spectral kernels to every frame of the window. [11] argues that applying a shared spectral response across rollout stages mismatches time-varying spectra in non-stationary systems, which motivates checking whether time conditioning improves our late coda. We did not test a time-conditioned spectral response, so this is an untested architectural candidate rather than an excluded one. Relatedly, [10] reports that per-frequency equal weighting can over-correct, which bears on any spectrally weighted objective built from our band decomposition; our present spectral loss term applies no frequency weighting at all.

One quoted number was withdrawn. An effective-rank figure circulated in an earlier draft as 2.81/7 has no artifact behind it; recomputation gives 2.71 under the entropy convention. Reproducible correlations of 0.9972 and 0.9975 establish strong co-variation with the frame term, but do not prove equality of gradients or dispensability of the spectral term.

10 Discussion and conclusion

The strongest result is the localization of the measured accuracy deficit. In the anchor training panel, the late third carries 64.1% of the mean per-record Marmousi error share, despite retaining substantial signal energy. The time decomposition, conditional spectral counterfactuals, and failed long intervention together show why an aggregate objective can improve without resolving the relevant prediction error. These diagnostics form a reusable experiment-selection procedure; they are not evidence of a universal error floor.

The present operator demonstrates a conditional query-time advantage while leaving a clear accuracy gap. Standard decoding is $2.5\text{--}5.1\times$ faster than the two tested classical configurations, but its Marmousi median error is $3.3\text{--}6.9\times$ larger, and the cheapest tested classical configuration is more accurate than the operator on every family median in Table 2. The comparisons of §4 add texture without changing this account: on train-development records the coarse-grid classical error is dispersive (phase lag, spectral downshift) where the neural error is an in-phase late-coda amplitude deficit, but the coarse solve still beats the operator on the layered record, and the equal-budget DeepONet baseline collapsed rather than providing a converged comparison point. Uniform training-panel accuracy is below 0.1, whereas layered and Marmousi full-window errors remain 0.2303 and 0.3965. These values describe the evaluated checkpoint and panels. Clean held-out performance still requires translated-crop grouping and an untouched test protocol.

The diagnostics prioritize three experiments. First, broaden temporal sampling support and test transfer on an independently fixed time grid. Second, supervise the correction against an explicit detached residual target and evaluate whether its direction improves. Third, compare a numerical coarse solve with a learned discrepancy correction, accounting for medium restriction, temporal substeps, free-surface treatment, and absorbing-boundary states. A short-window rollout with propagated state is another candidate, but must be trained and evaluated for accumulated rollout error. None of these routes is established by the current evidence. Their value should be judged by reproducible same-protocol improvements in complex-media late-time error, without degrading other families or obscuring total query cost.

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